

PentTESTGORage

john

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🏷 tags	john-the-ripper zip-password-cracking

We are given a zip file

```
> file john.zip
john.zip: Zip archive data, at least v2.0 to extract
```

when we try to extract the zip file it asks for password

```
> 7z x john.zip

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,4 CPUs Intel(R)
Core(TM) i5-7200U CPU @ 2.50GHz (806E9),ASM,AES-NI)

Scanning the drive for archives:
1 file, 205 bytes (1 KiB)

Extracting archive: john.zip
--
Path = john.zip
Type = zip
Physical Size = 205

Enter password (will not be echoed):
ERROR: Wrong password : flag.txt

Sub items Errors: 1

Archives with Errors: 1

Sub items Errors: 1
```

We don't know the password of the zip file but we can crack it using the john-the-ripper utility.

For that first we need to extract the hash from the zip file which can be done using a utility called `zip2john` which comes with the john-the-ripper.

```
> zip2john john.zip | tee hash.txt
ver 78.8 john.zip/flag.txt PKZIP Encr: cmplen=55, decmplen=43, crc=F0FE6A03
john.zip/flag.txt:$pkzip2$1*1*2*0*37*2b*f0fe6a03*0*26*0*37*f0fe*404e*1bb9ed2733f84bc4
e09bf52edcd11bf19d765e1e8226a3217f93c1ad4fbd68da9748423667f6e8f3f425d0d7f79ef8c651d2d
2ec0b650e*$/pkzip2$:flag.txt:john.zip::john.zip
```

now we can use `john-the-ripper` to crack the password using `rockyou.txt` wordlist.

```
> john hash.txt --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
eminem          (john.zip/flag.txt)
1g 0:00:00:00 DONE (2021-01-16 09:55) 33.33g/s 273066p/s 273066c/s 273066C/s 123456..
whitetiger
Use the "--show" option to display all of the cracked passwords reliably
Session completed
```

we were able to crack the password as `eminem` which we can use to extract the flag file.

```
> 7z x john.zip -peminem

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Core(TM) i5-7200U CPU @ 2.50GHz (806E9),ASM,AES-NI)

Scanning the drive for archives:
1 file, 205 bytes (1 KiB)

Extracting archive: john.zip
--
Path = john.zip
Type = zip
Physical Size = 205

Everything is Ok

Size:          43
Compressed: 205
```

```
> cat flag.txt  
ctf{cr4ck!ng_p4ssw0rds_is_345y_w!th_j0hn}
```



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